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Professor Patrick Maher
MIS545.EA: Applied Data Mining
Assignment 4

Taxi Cancellation Prediction Report

1. Objective

The objective of this project was to develop and compare multiple machine learning models to predict taxi booking cancellations using the dataset `taxi-cancellation-case.csv`, implemented through the script `case5.R`.

2. Required Packages

The following R packages were required for this project:

- `adabag`
- `base`
- `caret`
- `datasets`
- `doParallel`
- `foreach`
- `gains`
- `ggplot2`
- `graphics`
- `grDevices`
- `iterators`
- `lattice`
- `methods`
- `parallel`
- `rpart`
- `rpart.plot`
- `stats`
- `utils`

These packages were used for data manipulation, visualization, model building, and evaluation.

3. Data Preparation

The dataset **taxi-cancellation-case.csv** was loaded and preprocessed to ensure it was suitable for modeling.

3.1 Handling Missing Values

All missing values (NA) were replaced with 0:

```
taxi.df[is.na(taxi.df)] <- 0
```

3.2 Converting Variables to Factors

Categorical variables were converted into factors:

```
taxi.df$vehicle_model_id <- as.factor(taxi.df$vehicle_model_id)
taxi.df$package_id <- as.factor(taxi.df$package_id)
taxi.df$travel_type_id <- as.factor(taxi.df$travel_type_id)
```

3.3 Feature Engineering

Date variables were transformed into more meaningful components:

- Day of the week
- Hour of the day

Additionally, a new feature `trip_length` was computed using geographic coordinates:

```
taxi.df$trip_length <- dist(taxi.df$from_long, taxi.df$from_lat,
                           taxi.df$to_long, taxi.df$to_lat)
```

4. Data Splitting

The dataset was divided into training and validation sets:

```
set.seed(1)
train.ind <- sample(1:dim(taxi.df)[1], 0.6*dim(taxi.df)[1])
train.df <- taxi.df[train.ind, selected.vars]
valid.df <- taxi.df[-train.ind, selected.vars]
```

- 60% Training Data
- 40% Validation Data

5. Models Implemented

Four machine learning models were trained and evaluated:

5.1 K-Nearest Neighbors (KNN)

- Implemented using the caret package
- Tuned with $k = 1$ and $k = 10$

5.2 Decision Tree

- Built using the rpart package
- Pruned using cross-validation error to avoid overfitting

5.3 Boosting

- Implemented using the adabag package
- Combines multiple weak learners to improve prediction performance

5.4 Logistic Regression

- Implemented using glm
- Required adjustments to factor levels between training and validation datasets

6. Model Evaluation

All models were evaluated using:

- Confusion Matrix
- Accuracy
- Gains Chart

7. Results

Model	Accuracy	Interpretation
KNN	92.12%	High accuracy but poor detection of cancellations
Decision Tree	92.12%	Predicted mostly majority class
Boosting	92.18%	Best overall performance
Logistic Regression	29.80%	Poor performance due to instability

8. Key Observations

- The dataset is **highly imbalanced**, with most cases being non-cancellations
- High accuracy is misleading because models tend to predict the majority class
- KNN and Decision Tree showed limited ability to detect cancellations
- Boosting performed better by capturing more patterns in the data
- Logistic regression performed poorly due to:
 - Rank-deficiency issues
 - Factor level inconsistencies

9. Conclusion

Among all models, **Boosting provided the best performance** with an accuracy of **92.18%**, while also offering better detection of cancellation cases compared to other models.

Although KNN and Decision Tree achieved similar accuracy, they were biased toward the majority class and did not effectively identify cancellations.

Logistic regression was not suitable for this dataset due to instability and poor predictive performance.

10. Limitations

- Strong class imbalance affects model reliability
- Replacing missing values with 0 may introduce bias
- Some predictors may be redundant, impacting model stability

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case5.R boost gain train.valid rt knnfit pft taxi

Name

Type

Value

boost

list [9] (S3: boosting)

List of length 9

formula

Car_Cancellation.factor ~ .

trees

list [100]

List of length 100

weights

double [100]

1.277 0.742 0.466 0.406 0.300 0.233 ...

votes

double [6000 x 2]

14.68 14.63 13.22 10.28 10.68 11.09 1.57 1.61 3.03 5.96 5.57 5.15 ...

prob

double [6000 x 2]

0.9033 0.9008 0.8135 0.6329 0.6572 0.6829 0.0967 0.0992 0.1865 0.3671 0.3428 0.3 ...

class

character [6000]

0 0 0 0 0 0 ...

importance

list [13]

0.00 11.53 1.91 8.49 8.41 3.22 ...

terms

formula

Car_Cancellation.factor ~ vehicle_model_id + package_id + travel_type_id + ...

call

language

boosting(formula = Car_Cancellation.factor ~ ., data = train.df[, -14])

(No selection)

Console Terminal Background Jobs

R - R 4.5.3 - /cloud/project/

Warning: lower distance predictions values train groups requested

> plot(c(0, gain\$auc.pct, of total sum(valid.df\$Car_Cancellation)) <- (0, gain\$auc.obs),

+ xlab="x cases", ylab="Cumulative", main="", type="l", col = "grey")

> lines(c(0, sum(valid.df\$Car_Cancellation)) <- (0, sum(valid.df\$[13]), lty=2)

> confusionMatrix(as.factor(false.pred) ~ 0, 1, 0)

+ as.factor(valid.df\$Car_Cancellation))

Confusion Matrix and Statistics

Prediction 0 1

0 3683 313

1 2 2

Accuracy : 0.9212

95% CI : (0.9125, 0.9294)

No Information Rate : 0.9212

P-Value [Acc > NRI] : 0.315

Kappa : 0.806

McNemar's Test P-Value : <2e-16

Sensitivity : 0.999457

Specificity : 0.999457

Environment History Connections Tutorial

R - Global Environment

Data

boost

Large boosting (9 elements, 51.9 MB)

gain

List of 15

knnfit

Large train (25 elements, 4.6 MB)

pft

List of 15

reg

Large glm (30 elements, 6.3 MB)

rt

List of 15

taxi.df

10000 obs. of 23 variables

train.df

6000 obs. of 14 variables

valid.df

4000 obs. of 14 variables

Files Plots Packages Help Viewer Presentation

Install Update

Package

Description

Source

default/latest

Version

S7

An Object Oriented System Meant to Be

PPM (default/latest)

0.2.2

scales

Scale Functions for Visualization

PPM (default/latest)

1.4.0

shape

Functions for Plotting Graphical Shapes

PPM (default/latest)

1.4.6.1

sparsevctrs

Sparse Vectors for Use in Data Frames

PPM (default/latest)

0.3.6

spatial

Functions for Kriging and Point Pattern

CRAN

7.3-18

splines

Regression Spline Functions and Classes

Base

4.5.3

SQUAREM

Squared Extrapolation Methods for Acc

PPM (default/latest)

2026.1

stats

The R Stats Package

Base

4.5.3

stats4

Statistical Functions using S4 Classes

Base

4.5.3

stringi

Fast and Portable Character String Proce

PPM (default/latest)

1.8.7

stringr

Simple, Consistent Wrappers for Comm

PPM (default/latest)

1.6.0

survival

Survival Analysis

CRAN

3.8-6

tccltk

Tcl/Tk Interface

Base

4.5.3

tibble

Simple Data Frames

PPM (default/latest)

3.3.1

tidyr

Tidy Messy Data

PPM (default/latest)

1.3.2

tidyselect

Select from a Set of Strings

PPM (default/latest)

1.2.1

timechange

Efficient Manipulation of Date-Times

PPM (default/latest)

0.4.0

timeDate

Rmetrics - Chronological and Calendar

PPM (default/latest)

4052.112

tools

Tools for Package Development

Base

4.5.3

tzdb

Time Zone Database Information

PPM (default/latest)

0.5.0

utils

Unicode Text Processing

PPM (default/latest)

1.2.6

utils

The R Utils Package

Base

4.5.3

vctrs

Vector Helpers

PPM (default/latest)

0.7.3

viridisLite

Colorblind-Friendly Color Maps (Lite Ver

PPM (default/latest)

0.4.3

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case5.R boost gain train.valid rt knnfit pft taxi

Name

Type

Value

knnfit

list [25] (S3: train, train.forms

List of length 25

method

character [1]

'knn'

modelInfo

list [13]

List of length 13

modelType

character [1]

'Classification'

results

list [2 x 5] (S3: data.frame)

A data.frame with 2 rows and 5 columns

pred

NULL

Pairlist of length 0

bestTune

list [1 x 1] (S3: data.frame)

A data.frame with 1 row and 1 column

call

language

train.formulaInfo ~ as.factor(Car_Cancellation) ~ ., data = train.df, meth ...

dots

list [0]

List of length 0

metric

character [1]

'Accuracy'

control

list [27]

List of length 27

finalModel

list [8] (S3: knn3)

List of length 8

preProcess

NULL

Pairlist of length 0

trainingData

list [6000 x 14] (S3: data.frame)

A data.frame with 6000 rows and 14 columns

ptype

list [0 x 13] (S3: data.frame)

A data.frame with 0 rows and 13 columns

resample

list [25 x 3] (S3: data.frame)

A data.frame with 25 rows and 3 columns

resampledCM

list [50 x 6] (S3: data.frame)

A data.frame with 50 rows and 6 columns

perNames

character [2]

'Accuracy' 'Kappa'

maximize

logical [1]

TRUE

ylimits

NULL

Pairlist of length 0

times

list [3]

List of length 3

levels

character [2]

0 1

coefnames

formula

as.factor(Car_Cancellation) ~ vehicle_model_id + package_id + travel_type_id + ...

contrasts

list [5]

List of length 5

levels

list [5]

List of length 5

(No selection)

Console Terminal Background Jobs

R - R 4.5.3 - /cloud/project/

> View(gpft)

> View(knnfit)

> View(gpft)

> View(gpft)

Environment History Connections Tutorial

R - Global Environment

Data

boost

Large boosting (9 elements, 51.9 MB)

gain

List of 15

knnfit

Large train (25 elements, 4.6 MB)

pft

List of 15

reg

Large glm (30 elements, 6.3 MB)

rt

List of 15

taxi.df

10000 obs. of 23 variables

train.df

6000 obs. of 14 variables

valid.df

4000 obs. of 14 variables

Values

pred

Large numeric (4000 elements, 560.1 MB)

selected.vars

num [1:14] 3 4 5 6 7 8 9 12 13 20 ...

train.ind

int [1:6000] 1817 8004 4775 9725 8462 4050 8789 1301 8522 1799 ...

Functions

dist

function (long1, lat1, long2, lat2)

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Install Update

Package

Description

Source

default/latest

Version

S7

An Object Oriented System Meant to Be

PPM (default/latest)

0.2.2

scales

Scale Functions for Visualization

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Sparse Vectors for Use in Data Frames

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Functions for Kriging and Point Pattern

CRAN

7.3-18

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Regression Spline Functions and Classes

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Squared Extrapolation Methods for Acc

PPM (default/latest)

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Statistical Functions using S4 Classes

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PPM (default/latest)

1.2.1

timechange

Efficient Manipulation of Date-Times

PPM (default/latest)

0.4.0

timeDate

Rmetrics - Chronological and Calendar

PPM (default/latest)

4052.112

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File Edit Code View Plots Session Build Debug Profile Tools Help

case5.R boost gain train.df valid.df rt knnFit pft taxi.df

Name Type Value

pft	list [15] (53: rpart)	List of length 15
frame	list [9 x 8] (53: data.frame)	A data.frame with 9 rows and 8 columns
where	integer [6000]	2 2 4 2 8 2 ...
call	language	rpart(formula = Car_Cancellation ~., data = train.df)
terms	formula	Car_Cancellation ~ vehicle_model_id + package_id + travel_type_id + from_ar ...
cptable	double [4 x 5]	0.0235 0.0231 0.0119 0.0100 1.0000 1.0000 2.0000 4.0000 1.0000 0.9765 0.9533 0.9 ...
method	character [1]	'anova'
parms	NULL	Pairlist of length 0
control	list [9]	List of length 9
functions	list [2]	List of length 2
numresp	integer [1]	1
splits	double [40 x 5]	6.00e+03 6.00e+03 6.00e+03 6.00e+03 6.00e+03 0.00e+00 -1.00e+00 1.00e+00 ...
csplit	integer [12 x 20]	1 1 1 2 1 3 1 1 1 3 1 3 1 1 3 1 1 3 1 1 1 1 1 2 1 1 1 3 1 2 1 1 1 1 1 1 ...
variable.importance	double [11]	9.51 9.36 6.63 5.39 4.55 1.94 ...
y	integer [6000]	0 0 0 0 0 ...
ordered	logical [13]	FALSE FALSE FALSE FALSE FALSE ...

(No selection)

Console Terminal Background Jobs

R 4.5.3 - /cloud/project/

Reference

Prediction 0 1
0 940 63
1 2745 252

Accuracy : 0.298
95% CI : (0.2839, 0.3124)
No Information Rate : 0.9212
P-Value [Acc > NRI] : 1

Kappa : 0.0113

McNemar's Test P-Value : <2e-16

Sensitivity : 0.25599
Specificity : 0.80000

Environment History Connections Tutorial

R - Global Environment - 568 MB

Data

- boost: Large boosting (9 elements, 51.9 MB)
- gain: List of 15
- knnFit: Large train (25 elements, 4.6 MB)
- pft: List of 15
- reg: Large glm (30 elements, 6.3 MB)
- rt: List of 15
- taxi.df: 10000 obs. of 23 variables
- train.df: 6000 obs. of 14 variables
- valid.df: 4000 obs. of 14 variables

Values

- pred: Large numeric (4000 elements, 560.1 kB)
- selected.vors: run [1:14] 3 4 5 6 7 8 9 12 13 20 ...
- train.ind: int [1:6000] 1017 8004 4775 9725 8462 4850 8789 1301 8522 1799 ...

Functions

- dist: function (lon1, lat1, lon2, lat2)

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Install Update

Package	Description	Source	default/latest	Version
S7	An Object Oriented System Meant to Be	PPM [default/latest]	0.2.2	
scales	Scale Functions for Visualization	PPM [default/latest]	1.4.0	
shape	Functions for Plotting Graphical Shapes	PPM [default/latest]	1.4.6.1	
sparsevectors	Sparse Vectors for Use in Data Frames	PPM [default/latest]	0.3.6	
spatial	Functions for Kriging and Point Pattern	CRAN	7.3-18	
splines	Regression Spline Functions and Classes	Base	4.5.3	
SQUAREM	Squared Extrapolation Methods for Acc	PPM [default/latest]	2026.1	
stats	The R Stats Package	Base	4.5.3	
stats4	Statistical Functions using S4 Classes	Base	4.5.3	
stringi	Fast and Portable Character String Proc	PPM [default/latest]	1.8.7	
stringr	Simple, Consistent Wrappers for Comm	PPM [default/latest]	1.6.0	
survival	Survival Analysis	CRAN	3.8-6	
tcclik	Tcl/TK Interface	Base	4.5.3	
tibble	Simple Data Frames	PPM [default/latest]	3.3.1	
tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2	
tidyselect	Select from a Set of Strings	PPM [default/latest]	1.2.1	
timechange	Efficient Manipulation of Date-Times	PPM [default/latest]	0.4.0	
timeDate	Rmetrics - Chronological and Calendar	PPM [default/latest]	4052.112	

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case5.R boost gain train.df valid.df rt knnFit pft taxi.df

Name Type Value

gain	list [15] (53: gains)	List of length 15
depth	double [90]	12 12 13 14 15 16 ...
obs	integer [90]	462 18 40 40 40 40 ...
cume.obs	integer [90]	462 480 520 560 600 640 ...
mean.resp	double [90]	0.0974 0.0556 0.0750 0.0750 0.1750 0.0750 ...
cume.mean.resp	double [90]	0.0974 0.0958 0.0942 0.0929 0.0983 0.0969 ...
cume.pct.of.total	double [90]	0.143 0.146 0.156 0.165 0.187 0.197 ...
lift	double [90]	124 71 95 95 222 95 ...
cume.lift	double [90]	124 122 120 118 125 123 ...
mean.prediction	double [90]	1 1 1 1 1 1 ...
min.prediction	double [90]	1 1 1 1 1 1 ...
max.prediction	double [90]	1 1 1 1 1 1 ...
conf	character [1]	'none'
optimal	logical [1]	FALSE
num.groups	double [1]	100
percents	logical [1]	FALSE

(No selection)

Console Terminal Background Jobs

R 4.5.3 - /cloud/project/

Session restored from your saved work on 2026-Apr-30 16:34:29 UTC (2 days ago)

> View(boost)
> View(knnFit)
> View(pft)
> View(reg)
> View(rpart)
> View(scales)
> View(shape)
> View(splines)
> View(SQUAREM)
> View(stats)
> View(stats4)
> View(stringi)
> View(stringr)
> View(survival)
> View(tcclik)
> View(tibble)
> View(tidyr)
> View(tidyselect)
> View(timechange)
> View(timeDate)
> View(tools)
> View(tzdb)
> View(unicode)
> View(units)
> View(vectors)
> View(viridisLite)

Environment History Connections Tutorial

R - Global Environment - 567 MB

Data

- boost: Large boosting (9 elements, 51.9 MB)
- gain: List of 15
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- pft: List of 15
- reg: Large glm (30 elements, 6.3 MB)
- rt: List of 15
- taxi.df: 10000 obs. of 23 variables
- train.df: 6000 obs. of 14 variables
- valid.df: 4000 obs. of 14 variables

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SQUAREM	Squared Extrapolation Methods for Acc	PPM [default/latest]	2026.1	
stats	The R Stats Package	Base	4.5.3	
stats4	Statistical Functions using S4 Classes	Base	4.5.3	
stringi	Fast and Portable Character String Proc	PPM [default/latest]	1.8.7	
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survival	Survival Analysis	CRAN	3.8-6	
tcclik	Tcl/TK Interface	Base	4.5.3	
tibble	Simple Data Frames	PPM [default/latest]	3.3.1	
tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2	
tidyselect	Select from a Set of Strings	PPM [default/latest]	1.2.1	
timechange	Efficient Manipulation of Date-Times	PPM [default/latest]	0.4.0	
timeDate	Rmetrics - Chronological and Calendar	PPM [default/latest]	4052.112	
tools	Tools for Package Development	Base	4.5.3	
tzdb	Time Zone Database Information	PPM [default/latest]	0.5.0	
unicode	Unicode Text Processing	PPM [default/latest]	1.2.6	
units	The R Units Package	Base	4.5.3	
vectors	Vector Helpers	PPM [default/latest]	0.7.3	
viridisLite	Colorblind-Friendly Color Maps (Lite Ve	PPM [default/latest]	0.4.3	

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case5.R

boost

gain

train.df

valid.df

rt

knmFit

pfit

reg

taxi.df

Show Attributes

Name	Type	Value
reg	list [30] (S3: glm, lm)	List of length 30
coefficients	double [48]	-19.379 -0.699 14.046 -0.468 -0.702 -0.737 ...
residuals	double [6000]	-1.00 -1.01 -1.07 -1.06 -1.15 -1.03 ...
fitted.values	double [6000]	0.00179 0.00538 0.06636 0.05561 0.12828 0.02621 ...
effects	double [6000]	40.55815 -0.01438 0.11541 0.00391 0.00975 0.02658 ...
R	double [48 x 48]	-1.91e+01 0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.00e+00 -4.82e-08 9.60e-04 ...
rank	integer [1]	41
qr	list [5] (S3: qr)	List of length 5
family	list [13] (S3: family)	List of length 13
linear.predictors	double [6000]	-6.32 -5.22 -2.64 -2.83 -1.92 -3.62 ...
deviance	double [1]	2663.67
aic	double [1]	2745.67
null.deviance	double [1]	3084.891
iter	integer [1]	16
weights	double [6000]	0.00179 0.00535 0.06196 0.05252 0.11183 0.02552 ...
prior.weights	double [6000]	1 1 1 1 1 ...
df.residual	integer [1]	5999
df.null	integer [1]	5999
y	double [6000]	0 0 0 0 0 ...
converged	logical [1]	TRUE
boundary	logical [1]	FALSE
model	list [6000 x 14] (S3: data.frame)	A data.frame with 6000 rows and 14 columns
call	language	glm(formula = Car_Cancellation ~., family = binomial, data = train.df)
formula	formula	Car_Cancellation ~.
terms	formula	Car_Cancellation ~ vehicle_model_id + package_id + travel_type_id + from_ar ...
data	list [6000 x 14] (S3: data.frame)	A data.frame with 6000 rows and 14 columns
offset	NULL	Pairlist of length 0
control	list [3]	List of length 3
method	character [1]	'glm.fit'
contrasts	list [1]	List of length 5

Console

Environment History Connections Tutorial

Import Dataset

568 MB

Global Environment

Data

boost

gain

knmFit

pfit

reg

rt

taxi.df

train.df

valid.df

Values

pred

selected.vars

train.ind

Functions

dist

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Install

Update

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tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2
tidyselect	Select from a Set of Strings	PPM [default/latest]	1.2.1
timechange	Efficient Manipulation of Date-Times	PPM [default/latest]	0.4.0
timeDate	Rmetrics - Chronological and Calendar	PPM [default/latest]	4052.112

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Show Attributes

Name	Type	Value
rt	list [15] (S3: rpart)	List of length 15
frame	list [9 x 8] (S3: data.frame)	A data.frame with 9 rows and 8 columns
where	integer [6000]	2 2 4 2 8 2 ...
call	language	rpart(formula = Car_Cancellation ~., data = train.df)
terms	formula	Car_Cancellation ~ vehicle_model_id + package_id + travel_type_id + from_ar ...
cptable	double [4 x 5]	0.0235 0.0231 0.0119 0.0100 0.0000 1.0000 2.0000 4.0000 1.0000 0.9765 0.9533 0.9 ...
method	character [1]	'anova'
parms	NULL	Pairlist of length 0
control	list [9]	List of length 9
functions	list [2]	List of length 2
numresp	integer [1]	1
splits	double [40 x 5]	6.00e+03 6.00e+03 6.00e+03 6.00e+03 6.00e+03 0.00e+00 -1.00e+00 1.00e+00 ...
csplit	integer [12 x 20]	1 1 1 2 1 3 1 1 1 1 1 3 1 1 3 1 1 3 1 ...
variable.importance	double [11]	9.51 9.36 6.63 5.39 4.55 1.94 ...
y	integer [6000]	0 0 0 0 0 ...
ordered	logical [13]	FALSE FALSE FALSE FALSE FALSE ...

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R 4.5.3 - /cloud/project/

> View(boost)

> View(gain)

> View(pfit)

> View(reg)

> View(rpart)

> View(selected.vars)

> View(train.ind)

> View(valid.df)

> View(boost)

> View(gain)

> View(pfit)

> View(reg)

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S7	An Object Oriented System Meant to Be	PPM [default/latest]	0.2.2
scales	Scale Functions for Visualization	PPM [default/latest]	1.4.0
shape	Functions for Plotting Graphical Shapes	PPM [default/latest]	1.4.6.1
sparsevctrs	Sparse Vectors for Use in Data Frames	PPM [default/latest]	0.3.6
spatial	Functions for Kriging and Point Pattern	CRAN	7.3-18
splines	Regression Spline Functions and Classes	Base	4.5.3
SQUAREM	Squared Extrapolation Methods for Accu	PPM [default/latest]	2026.1
stats	The R Stats Package	Base	4.5.3
stats4	Statistical Functions using S4 Classes	Base	4.5.3
stringi	Fast and Portable Character String Proc	PPM [default/latest]	1.8.7
stringr	Simple, Consistent Wrappers for Comm	PPM [default/latest]	1.6.0
survival	Survival Analysis	CRAN	3.8-6
tccl	Tcl/Tk Interface	Base	4.5.3
tibble	Simple Data Frames	PPM [default/latest]	3.3.1
tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2
tidyselect	Select from a Set of Strings	PPM [default/latest]	1.2.1
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timeDate	Rmetrics - Chronological and Calendar	PPM [default/latest]	4052.112

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package_id

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from_area_id

to_area_id

from_city_id

to_city_id

from

1

1

17712

12

0

2

1021

1323

0

0

13

2

2

17037

12

0

2

455

1930

0

0

13

3

3

761

12

0

2

814

393

0

0

13

4

4

868

12

0

2

297

212

0

0

13

5

5

21716

28

0

2

1237

330

0

0

13

6

6

38966

12

0

2

61

393

0

0

13

7

7

22196

12

0

2

409

1194

0

0

13

8

8

22200

12

0

2

1371

839

0

0

13

9

9

22201

12

1

3

1323

0

0

0

13

10

10

22202

12

0

2

1017

496

0

0

13

11

11

22203

12

0

2

1373

1390

0

0

13

12

12

819

12

0

2

1323

585

0

0

13

13

13

15846

12

2

3

142

0

0

0

13

14

14

3316

24

0

2

58

393

0

0

13

15

15

19469

85

0

2

393

392

0

0

13

16

16

15815

12

0

2

145

871

0

0

13

17

17

21903

12

0

2

1364

1053

0

0

13

18

18

7500

12

0

2

112

393

0

0

13

19

19

19322

12

0

2

515

186

0

0

13

20

20

12827

12

0

2

793

949

0

0

13

21

21

22223

12

0

2

455

1181

0

0

13

22

22

15611

12

0

2

516

393

0

0

13

23

23

21235

12

0

2

393

217

0

0

13

24

24

15529

12

2

3

217

0

0

0

13

25

25

22226

12

0

2

344

585

0

0

13

26

26

22229

12

2

3

1053

0

0

0

13

27

27

22230

85

0

2

1301

393

0

0

13

28

28

21333

28

0

2

1223

585

0

0

13

29

29

22234

12

0

2

58

1069

0

0

13

30

30

18777

12

1

3

293

0

0

0

13

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to_city_id

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mobile

1017

24

0

2

1150

393

0

0

0

0

8004

24

0

2

1070

934

15

0

0

0

4775

12

0

2

83

393

0

0

0

1

9725

12

0

2

293

1089

15

0

0

0

8462

12

0

2

1121

768

15

0

0

1

4050

12

0

2

471

168

0

0

0

0

8789

89

0

2

330

83

15

0

0

0

1301

12

0

2

1010

83

0

0

0

0

8522

65

0

2

625

168

15

0

0

0

1799

28

0

2

1267

1032

0

0

0

1

9941

12

0

2

585

1083

0

0

0

0

8229

12

0

2

1371

400

15

0

0

0

1129

12

0

2

122

1391

0

0

0

0

8921

12

0

2

293

1384

15

0

0

0

878

12

0

2

61

393

0

0

0

1

8677

12

0

2

542

1384

15

0

0

1

7845

65

0

2

393

585

15

0

0

1

5922

12

0

2

1089

393

15

0

0

0

6526

12

1

3

1315

0

0

0

0

0

5071

89

0

2

393

334

0

0

0

1

2900

85

2

3

488

0

0

0

0

1

7075

12

0

2

293

410

15

0

0

0

4650

12

0

2

1068

197

0

0

0

0

9638

24

0

2

1157

585

0

0

0

0

2159

12

0

2

455

1045

0

0

0

0

3476

12

0

2

1083

768

0

0

0

1

1948

12

0

2

396

393

0

0

0

1

2580

12

0

2

542

1150

0

0

0

0

1530

12

2

3

1327

0

0

0

0

0

7289

12

0

2

1063

1384

15

0

0

1

Showing 1 to 30 of 6,000 entries, 14 total columns

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pred	Large numeric (4000 elements, 560.1 kB)
- attr(*, "names")= chr [1:4000] "1" "10" "11" "12" ...	
- attr(*, "non-estim")= Named int [1:3994] 1 2 3 4 5 6 7 8 9 10 ...	
..- attr(*, "names")= chr [1:3994] "1" "10" "11" "12" ...	
selected.vars	num [1:14] 3 4 5 6 7 8 9 12 13 20 ...
train.ind	int [1:6000] 1017 8004 4775 9725 8462 4050 8789 1301 8522 1799 ...
Functions	
dist	function (long1, lat1, long2, lat2)

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case5.R boost gain train.df valid.df rt knnFit pft reg taxi.df

Filter

	vehicle_model_id	package_id	travel_type_id	from_area_id	to_area_id	from_city_id	to_city_id	online_booking	mobile
1	12	0	2	1021	1323	0	0	0	0
10	12	0	2	1017	496	0	0	0	0
11	12	0	2	1373	1190	0	0	0	0
12	12	0	2	1323	585	0	0	0	1
13	12	2	3	142	0	0	0	0	0
17	12	0	2	1384	1053	0	0	0	1
18	12	0	2	112	393	0	0	0	0
20	12	0	2	793	949	0	0	0	0
22	12	0	2	516	393	0	0	0	0
23	12	0	2	393	217	0	0	0	1
25	12	0	2	344	585	0	0	0	1
28	12	0	2	1223	585	0	0	0	0
31	12	0	2	1194	409	0	0	0	0
33	85	0	2	393	1056	0	0	0	1
35	12	0	2	479	393	0	0	0	0
37	12	0	2	393	1237	0	0	0	0
42	12	0	2	849	1312	0	0	0	1
43	12	0	2	392	297	0	0	0	1
47	12	0	2	572	793	0	0	0	0
49	85	0	2	410	393	0	0	0	0
50	65	0	2	450	393	0	0	0	1
52	12	0	2	1388	1190	0	0	0	1
60	12	0	2	393	1136	0	0	0	1
66	85	0	2	293	393	0	0	0	0
69	12	0	2	852	1078	0	0	0	0
71	12	0	2	1217	410	0	0	0	1
74	28	2	3	115	0	0	0	0	1
75	12	0	2	393	1031	0	0	0	1
76	12	0	2	293	1146	0	0	0	1
85	12	0	2	363	393	0	0	0	0

Showing 1 to 30 of 4,000 entries, 14 total columns

Console

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R - Global Environment

Data

- boost Large boosting (9 elements, 51.9 MB)
- gain List of 15
- knnFit Large train (25 elements, 4.6 MB)
- pft List of 15
- reg Large glm (30 elements, 6.3 MB)
- rt List of 15
- taxi.df 10000 obs. of 23 variables
- train.df 6000 obs. of 14 variables
- valid.df 4000 obs. of 14 variables

Values

- pred Large numeric (4000 elements, 560.1 kB)
- selected.vars num [1:14] 3 4 5 6 7 8 9 12 13 20 ...
- train.ind int [1:6000] 1017 8004 4775 9725 8462 4050 8789 1301 8522 1799 ...

Functions

- dist function (long1, lat1, long2, lat2)

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Package	Description	Source	default/latest	Version
S7	An Object Oriented System Meant to Be	PPM [default/latest]	0.2.2	
scales	Scale Functions for Visualization	PPM [default/latest]	1.4.0	
shape	Functions for Plotting Graphical Shapes	PPM [default/latest]	1.4.6.1	
sparsevctrs	Sparse Vectors for Use in Data Frames	PPM [default/latest]	0.3.6	
spatial	Functions for Kriging and Point Pattern	CRAN	7.3-18	
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SQUAREM	Squared Extrapolation Methods for Ace	PPM [default/latest]	2026.1	
stats	The R Stats Package	Base	4.5.3	
stats4	Statistical Functions using 54 Classes	Base	4.5.3	
strings	Fast and Portable Character String Proc	PPM [default/latest]	1.8.7	
stringr	Simple, Consistent Wrappers for Comm	PPM [default/latest]	1.6.0	
survival	Survival Analysis	CRAN	3.8-6	
tecltk	Tcl/Tk Interface	Base	4.5.3	
tidble	Simple Data Frames	PPM [default/latest]	3.3.1	
tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2	
tidyselect	Select from a Set of Strings	PPM [default/latest]	1.2.1	
timechange	Efficient Manipulation of Date-Times	PPM [default/latest]	0.4.0	
timeDate	Rmetrics - Chronological and Calendar	PPM [default/latest]	4052.112	

Function: dist (.GlobalEnv)

```

1 dist <- function (long1, lat1, long2, lat2){
2   rad <- pi/180
3   a1 <- lat1 * rad
4   a2 <- long1 * rad
5   b1 <- lat2 * rad
6   b2 <- long2 * rad
7   dlon <- b2 - a2
8   dlat <- b1 - a1
9   a <- (sin(dlat/2))^2 + cos(a1) * cos(b1) * (sin(dlon/2))^2
10  c <- 2 * atan2(sqrt(a), sqrt(1 - a))
11  R <- 6378.145
12  d <- R * c
13  return(d)
14 }
15

```